

Assignment 1

$$(a) \quad (113.015625)_{10} = (1110001.000001)_2$$

$$(b) \quad (5963)_7 = (1956)_{10} = (B76)_{13}$$

$$(c) \quad (593.0618)_{10} = (100111010111.0100101001011100)_{\text{Excess 4}}$$

$$(d) \quad (110101010101110101.0110101011)_2 = (6A4F3756)$$

$$(e) \quad (10110100110)_{1's} = -(01001011001)_2 = -(601)_{10}$$

$$(f) \quad (00110111101)_{2's} = +(00110111101)_2 = +(445)_{10}$$

$$(g) \quad (4563.0712)_8$$

$$= (100101110011.000111001010)_2$$

$$(h) \quad (1100001101101011010.01011001)_{\text{Excess 3}} \\ = (90387.26)_{10}$$

$$(i) \quad (10110100110)_{2's} = -(01001011010)_2 = -(602)_{10}$$

$$(j) \quad (77)_{10} = (205)_6$$

$$(k) \quad (100110.1101)_4 = (1044.31640625)_{10}$$

Question 2

$$\begin{array}{r}
 (2FB4)_{18} \\
 + (8B)_{18} \\
 \hline
 (2A1F)_{18}
 \end{array}$$

$$\begin{array}{r}
 \text{Base-18} \\
 2FB4 \\
 8B \\
 \hline
 2F2B
 \end{array}$$

$$\begin{array}{r}
 \text{Base 18} \\
 2FB4 \\
 8B \\
 \hline
 1D9F8 \\
 1A6HEX \\
 \hline
 16C9B8
 \end{array}$$

Question 3

Base-6

$$\begin{array}{r}
 112 \overline{) 101224} \quad 000503 \\
 \underline{0} \\
 10 \\
 \underline{0} \\
 101 \\
 \underline{0} \\
 1012 \\
 \underline{1009} \\
 42 \\
 \underline{0} \\
 424 \\
 \underline{340} \\
 44
 \end{array}$$

$$\begin{aligned}
 112 \times 1 &= 112 \\
 112 \times 2 &= 224 \\
 112 \times 3 &= 340 \\
 112 \times 4 &= 452 \\
 112 \times 5 &= 1004
 \end{aligned}$$

Quotient $(503)_6$
 Remainder $(44)_6$

Question 6

$$\begin{aligned}
 + (91)_{10} &= + (1011011)_2 \\
 &= + (01011011)_2 \\
 &= (01011011)_{2's} \\
 &\quad \begin{array}{r}
 11111 \\
 01011011 \\
 + 11011101 \\
 \hline
 100111000 \\
 \times
 \end{array} \\
 &\quad (00111000)_{2's}
 \end{aligned}$$

$$\begin{aligned}
 -35 &= - (100011)_2 \\
 &= - (00100011)_2 \\
 &= (11011101)_{2's}
 \end{aligned}$$

Adding two different sign numbers. So overflow not possible.

Assignment 2

Q1(a)

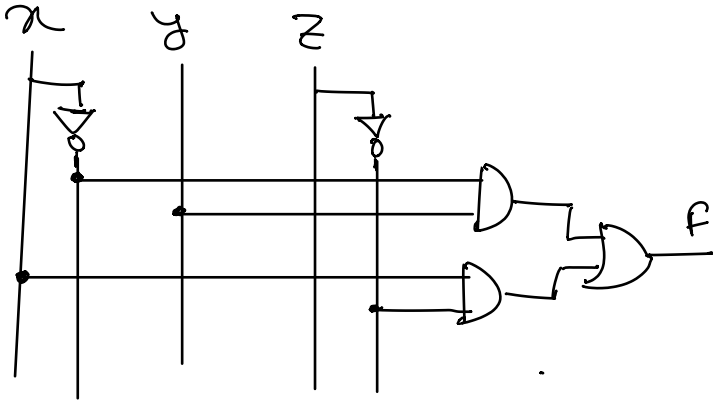
$$\begin{aligned} & (\overline{A}B + \overline{B}C) + \overline{(D + \overline{B}C)} \\ &= \overline{A}B + \overline{B}C \cdot \overline{D + \overline{B}C} \\ &= \overline{A}B \cdot \overline{B}C \cdot (D + \overline{B}C) \\ &= (\overline{A} + \overline{B}) (\overline{B} + \overline{C}) (D + \overline{B}C) \\ &= (A + \overline{B}) (B + \overline{C}) (D + \overline{B}C) \\ &= (A + \overline{B}) (BD + B\overline{B}C + \overline{C}D + \overline{B}C\overline{C}) \\ &= (A + \overline{B}) (BD + 0 + \overline{C}D + 0) \\ &= ABD + A\overline{C}D + \overline{B}BD + \overline{B}\overline{C}D \\ &= ABD + A\overline{C}D + \overline{B}\overline{C}D \end{aligned}$$

Q1(b)

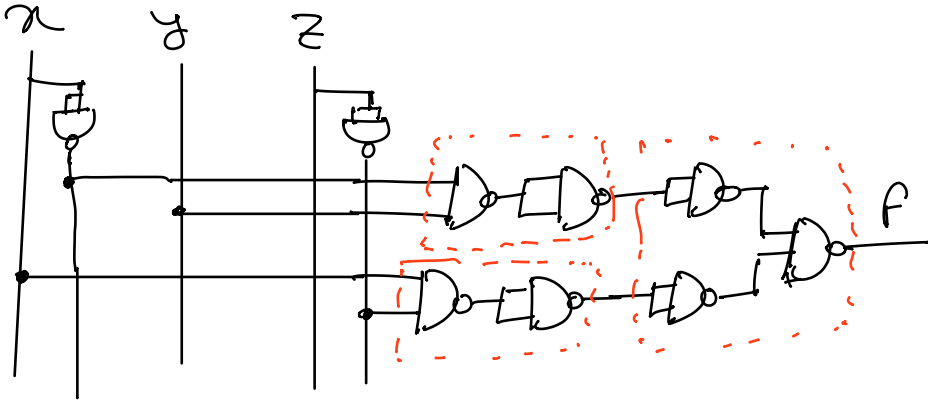
$$\begin{aligned} f(x, y, z) &= \pi(2, 6) \\ &= \pi(010, 110) \\ &= (x + y' + z) (x' + y' + z) \\ &= (y' + z + x x') \\ &= y' + z \end{aligned}$$

Ques 2

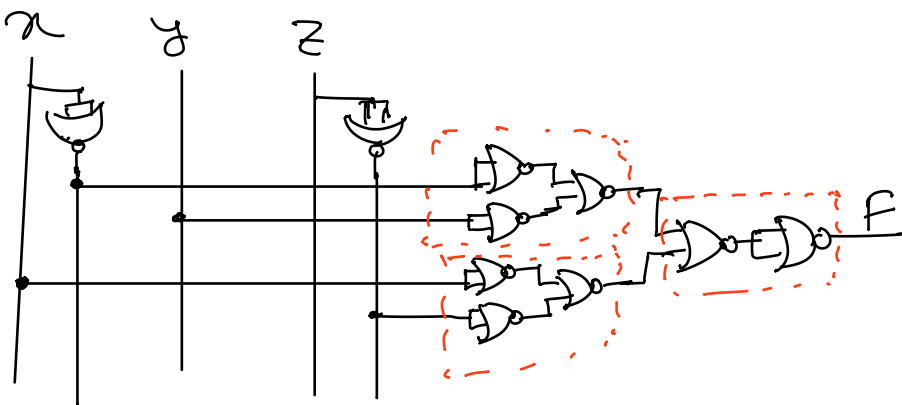
(a) $x'y + xz'$



Using NAND



Using NOR



Ques 3

$$(a) F(a, b, c)$$

$$= b + b'c(a+e)$$

$$= b + ab'c + b'ce'$$

$$= b + ab'c + 0$$

$$= (a+b)(b+b')(b+e)$$

$$= (a+b)(b+e)$$

$$= (a+b+ce')(aa'+b+e)$$

$$= (a+b+e)(a+b+e')(a+~~b+e~~)(a'+b+e)$$

$$= \Pi(000, 001, 100)$$

$$= \Pi(0, 1, 4)$$

$$(b) F(w, x, y, z)$$

$$= x'y + y'(z+x'z')$$

$$= x'y + y'z + x'y'z'$$

$$= ((x'y + y')(x'y + z)) + x'y'z'$$

$$= ((x'+y')(y+y')(\cancel{y+y'}) + x'y'z')$$

$$= ((x'+y')(x'+z)(y+z) + x'), ((x'+y')(x'+z)(y+z) + y').$$
$$((x'+y')(x'+z)(y+z) + z')$$

$$= (x'+y'+x')(x'+z+x')(y+z+x')$$

$$(x'+y'+y')(x'+z+y')(y+z+y')$$

$$(x'+y'+z')(x'+z+z')(y+z+z')$$

$$= (n'+y') (n'+z) (n'+y+z) \cancel{(n'+y')} (n'+y'+z) \\ (n'+y'+z')$$

$$= (n'+y'+zz') (n'+yy'+z) (n'+y+z) (n'+y'+z) \\ (n'+y'+z')$$

$$= (n'+y'+z) (n'+y'+z') (n'+y+z) \cancel{(n'+y'+z)} \\ \cancel{(n'+y+z)} \cancel{(n'+y'+z)} \cancel{(n'+y'+z')}$$

$$= (\omega\omega' + n'+y'+z) (\omega\omega' + n'+y'+z') + \\ (\omega\omega' + n'+y+z) (\omega\omega' + n'+y+z')$$

$$= (\omega + n'+y'+z) (\omega' + n'+y'+z) (\omega + n'+y'+z') \\ (\omega' + n'+y'+z') (\omega + n'+y+z) (\omega' + n'+y+z)$$

$$= \pi(0110, 1110, 0111, 1111, 0100, 1100)$$

$$= \pi(6, 14, 7, 15, 4, 12)$$

$$= \pi(4, 6, 7, 12, 14, 15)$$

Ques 4

$$\begin{aligned} & ((a+b+c')(a+b)(b'+c'))' \\ &= (a+b+c')' + (a+b)' + (b'+c')' \\ &= a'b'c + a'b' + bc \\ &= a'b'(c+1) + bc \\ &= a'b' + bc \end{aligned}$$

Ques 5

$$\begin{aligned} & F(P, Q, R, S) \\ &= P + Q'RS' \\ &= P(Q+Q') + Q'RS' \\ &= PQ + PQ' + Q'RS' \\ &= PQ(R+R') + PQ'(R+R') + Q'RS' \\ &= PQR + PQR' + PQ'R + PQ'R' + Q'RS' \\ &= PQR(S+S') + PQR'(S+S') + PQ'R(S+S') + PQ'R'(S+S') \\ &\quad + (P+P')Q'RS' \\ &= PQR S + PQR S' + PQR' S + PQR' S' + PQ'R S + PQ'R S' \\ &\quad + PQ'R' S + PQ'R' S' + \cancel{PQ'RS'} + P'Q'RS' \\ &= \Sigma(1111, 1110, 1101, 1100, 1011, 1010, 1001, \\ &\quad 1000, 0010) \\ &= \Sigma(15, 14, 13, 12, 11, 10, 9, 8, 2) \\ &= \Sigma(2, 8, 9, 10, 11, 12, 13, 14, 15) \end{aligned}$$